Sound Waves:
Characteristics and Applications

Important Questions with Answers

Class 9 Science | NCERT Chapter 10 | 42 Questions

MCQ · True/False · Fill in the blanks · Short answer · Numericals · Long answer

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Tip: cover the green answer box, try the question first, then check. Compare with your own textbook wording.

A. Multiple Choice Questions

Q1 - Q5

Q1. Sound is produced by:

(a) stationary objects (b) vibrating objects (c) light waves (d) magnetic fields

Ans: (b) Vibrating objects. Sound stops when the vibration stops.

Q2. Sound cannot travel through:

(a) steel (b) water (c) air (d) vacuum

Ans: (d) Vacuum. Sound needs a material medium (solid, liquid or gas).

Q3. Sound waves in air are:

(a) transverse (b) longitudinal (c) electromagnetic (d) stationary

Ans: (b) Longitudinal. Particles vibrate parallel to the direction of propagation.

Q4. If 20 compressions pass a point in 4 seconds, the frequency is:

(a) 80 Hz (b) 5 Hz (c) 10 Hz (d) 0.2 Hz

Ans: (b) 5 Hz. Frequency = number of oscillations ÷ time = 20 ÷ 4 = 5 Hz.

Q5. The speed of sound is highest in:

(a) air (b) water (c) steel (d) vacuum

Ans: (c) Steel. Sound travels fastest in solids (about 5000 m s^{-1}), slower in liquids, slowest in gases.

A. Multiple Choice Questions (contd.)

Q6. The audible range of human hearing is:

(a) below 20 Hz (b) 20 Hz to 20 kHz (c) above 20 kHz (d) above 1 MHz

Ans: (b) 20 Hz to 20 kHz. Below this is infrasonic and above it is ultrasonic.

Q7. Taking the speed of sound as 340 m s^{-1}, the minimum distance of a reflecting surface to hear an echo is:

(a) 8.5 m (b) 17 m (c) 34 m (d) 68 m

Ans: (b) 17 m. Sound must travel to the wall and back in 0.1 s: 340 × 0.1 = 34 m, so the wall is 17 m away.

Q8. Bats locate their prey in the dark using:

(a) infrasonic waves (b) ultrasonic waves (c) light (d) audible sound

Ans: (b) Ultrasonic waves. Bats send ultrasonic pulses and sense the reflected echoes (echolocation).

B. True or False

Q9 - Q14

Q9. Sound is produced by vibrating objects.

Ans: True. Vibration is the periodic to-and-fro motion of an object, and it produces sound.

Q10. Sound can travel through a vacuum.

Ans: False. Sound needs a material medium; it cannot propagate in vacuum.

Q11. The particles of the medium travel with the sound wave from the source to the listener.

Ans: False. The particles only vibrate about their mean positions; the disturbance and energy travel.

Q12. In a compression, the density of the medium is higher than the average density.

Ans: True. A compression is a region of higher density; a rarefaction is a region of lower density.

Q13. Wavelength is the distance between two consecutive crests or two consecutive troughs.

Ans: True. Its SI unit is the metre (m).

Q14. Frequency and time period of a wave are directly proportional.

Ans: False. They are inversely related: $\nu = 1 \div T$.

B. True or False (contd.)

Q15. The speed of sound in air increases when the temperature increases.

Ans: True. For example, about 331 m s^{-1} at 0 °C and 344 m s^{-1} at 22 °C.

Q16. Pitch and loudness mean the same thing.

Ans: False. Pitch is how we perceive frequency; loudness is how we perceive amplitude.

Q17. Reverberation is caused by multiple reflections arriving less than 0.05 s apart.

Ans: True. The sound persists after the source stops emitting.

Q18. Ultrasonic waves have frequencies below 20 Hz.

Ans: False. Ultrasonic waves have frequencies above 20 kHz; below 20 Hz is infrasonic.

C. Fill in the Blanks

Q19 - Q24

Q19. The object that produces sound is called the _____ of the sound.

Ans: source.

Q20. The SI unit of frequency is _____.

Ans: hertz (Hz), or per second (s^{-1}).

Q21. Speed of sound = _____ × _____.

Ans: wavelength × frequency ($v = \lambda \times \nu$).

Q22. In a sound wave, a region of higher density is called a _____ and a region of lower density is called a _____.

Ans: compression; rarefaction.

Q23. Sound waves with frequency above 20 kHz are called _____ waves.

Ans: ultrasonic.

Q24. On the density-distance graph of a sound wave, the highest point is the _____ and the lowest point is the _____.

Ans: crest; trough.

D. Short Answer Questions

Q25 - Q29

Q25. What is vibration? How does the rubber band activity show that sound is produced by vibration?

Ans: Vibration is the periodic to-and-fro motion (oscillation) of an object. A plucked rubber band on a box vibrates and produces sound; when it stops vibrating, the sound stops.

Q26. Why can astronauts on a spacewalk not hear each other directly?

Ans: Outer space is a near vacuum with no medium, and sound cannot propagate without a medium. They communicate through special devices fitted into their spacesuits.

Q27. Describe the vacuum bell jar experiment and state its conclusion.

Ans: An electric bell is kept in a bell jar connected to a vacuum pump. As air is pumped out, the sound becomes fainter and almost disappears, though the bell is seen ringing. When air is let back, the sound returns. So sound needs a medium to propagate.

Q28. Define compression and rarefaction.

Ans: A compression is a region of the medium where the density is higher than average; a rarefaction is a region where the density is lower than average. They are produced alternately by a vibrating source.

Q29. What is a longitudinal wave? Give one example.

Ans: A wave in which the particles of the medium vibrate parallel to the direction of wave propagation. A sound wave in air is an example.

D. Short Answer Questions (contd.)

Q30. Define wavelength, frequency and time period. How are frequency and time period related?

Ans: Wavelength: distance between two consecutive crests or troughs (m). Frequency: number of density oscillations per second at a point (Hz). Time period: time for one complete oscillation (s). They are related by $\nu = 1 \div T$.

Q31. Distinguish between pitch and loudness.

Ans: Pitch is how humans perceive frequency: high frequency sounds are shrill, low frequency sounds are deep. Loudness is how humans perceive amplitude: larger amplitude sounds louder. Loudness is measured in decibels (dB).

Q32. Differentiate between an echo and reverberation.

Ans: An echo is a reflected sound heard separately, when the gap between the original and reflected sound is at least 0.1 s. Reverberation is the persistence of sound due to repeated reflections arriving less than 0.05 s apart, as in a large hall.

Q33. Name two applications of ultrasonic waves.

Ans: Ultrasonography (imaging internal organs without surgery) and breaking kidney stones into smaller pieces. Others: ultrasonic welding and cleaning, detecting defects in metal blocks.

Q34. What is echolocation? Name two animals that use it.

Ans: Echolocation is locating objects using reflected sound waves. Bats emit short bursts of ultrasonic waves and sense the echoes to find obstacles and prey. Dolphins and whales also use it.

E. Numericals

Q35 - Q37

Q35. A sound wave in air has a frequency of 50 Hz and a wavelength of 6.8 m. Find its speed and time period.

Ans: Speed $v = \lambda \times \nu = 6.8\ m \times 50\ Hz = 340\ m\ s^{-1}$.
Time period $T = 1 \div \nu = 1 \div 50 = 0.02\ s$.

Q36. A person claps in a large hall and hears an echo after 1.2 s. If the speed of sound is 340 m s^{-1}, how far is the reflecting wall?

Ans: Total distance travelled = $v \times t = 340 \times 1.2 = 408\ m$ (to the wall and back).
Distance of the wall = $408 \div 2 = 204\ m$.

Q37. A sonar signal sent from a ship returns after 3.2 s. The speed of sound in sea water is 1500 m s^{-1}. Find the depth of the sea.

Ans: Time to reach the sea bed = $3.2 \div 2 = 1.6\ s$.
Depth = speed × time = $1500 \times 1.6 = 2400\ m$.

F. Long Answer Questions

Q38 - Q40

Q38. Describe an experiment to show that sound needs a medium to propagate. What does it tell us about sound in space?

Ans: Vacuum bell jar: an electric bell is switched on inside a bell jar connected to a vacuum pump, and its loudness is noted.
• As air is sucked out, the sound becomes fainter; at near vacuum, almost no sound is heard though the bell is seen ringing.
• When air is let back in, the sound gradually becomes as loud as before.
Conclusion: sound cannot propagate in vacuum; it needs a solid, liquid or gas as medium. Space is a near vacuum, so astronauts cannot hear each other directly and use special communication devices.

Q39. Using a piston in a tube, explain how a sound wave is produced and travels through air. Define a sound wave.

Ans: A piston oscillates at one end of an air-filled tube. When it moves forward it pushes nearby air particles, creating a compression (higher density). When it moves backward, a rarefaction (lower density) forms.
The compressed particles collide with neighbouring particles and pass the compression forward; the same happens for the rarefaction. Alternate compressions and rarefactions travel away from the source.
The particles only oscillate about their mean positions; they do not travel with the wave. A sound wave is the disturbance of alternating compressions and rarefactions propagating through a medium without the actual flow of the medium’s particles.

Q40. Explain the graphical representation of a sound wave. What are crest, trough, wavelength and amplitude? How do waves of low and high amplitude differ?

Ans: The density of the medium at an instant is plotted on the y-axis against distance on the x-axis; the average density is a dashed line.
• Crest: highest point, maximum density (compression). Trough: lowest point, minimum density (rarefaction).
• Wavelength: distance between two consecutive crests or troughs.
• Amplitude: maximum change in density from the average.
A high amplitude wave has taller crests and deeper troughs, carries more energy and sounds louder; a low amplitude wave carries less energy and sounds softer.

F. Long Answer Questions (contd.)

Q41. Derive the relation $v = \lambda \times \nu$. State the factors that affect the speed of sound, with values for steel, water and air.

Ans: Speed = distance ÷ time. In one time period T the wave travels one wavelength λ, so $v = \lambda \div T$.
Since frequency $\nu = 1 \div T$, $v = \lambda \times \nu$ (speed = wavelength × frequency).
The speed depends on the medium: fastest in solids, slower in liquids, slowest in gases (steel about 5000 m s^{-1}, water about 1500 m s^{-1}, air about 340 m s^{-1}). In air it also rises with temperature and humidity (331 m s^{-1} at 0 °C, 344 m s^{-1} at 22 °C).

Q42. Explain echo, reverberation and sonar with the formulas used.

Ans: • Echo: reflected sound heard separately if the gap is at least 0.1 s. In 0.1 s sound travels 340 × 0.1 = 34 m (to the wall and back), so the minimum distance is 17 m. Distance from wall = v × t ÷ 2.
• Reverberation: repeated reflections in a large hall arriving less than 0.05 s apart make sound persist; soft panels and curtains reduce it.
• Sonar: ultrasonic waves are sent into water and the reflected waves are analysed to find the distance, direction and speed of objects such as submarines. Distance = v × t ÷ 2 (e.g. 0.90 s at 1530 m s^{-1} gives 688.5 m).